Option A (revised) — Stage card

Stage 3 — Basic Engineering Gate: Critical Design Review (CDR)

The 60% design deliverables are completed and the cost and schedule are baselined. This is when the bulk of controls engineering begins.

Required Controls Inputs

- Preliminary Design Package
- Developing P&IDs
- Preliminary layout

Controls Outputs

- Preliminary I/O list
- Panel concept
- Long-lead orders placed

Controls Work Performed

- Build a preliminary I/O count and list from the P&IDs.
- Decide central vs. remote panels.
- Order long-lead items once approved.

Relevant Engineering Deliverables

- P&ID (ready for PHA)
- EBOM (about 80% ready)
- Equipment specifications
- Preliminary Interlock Matrix
- Preliminary Control Narrative
- Preliminary electrical load list
- I/O list
- Updated layout and utility requirements
- Preliminary RFPs and bid documents
- Class 3 cost estimate and Level 3 schedule
- Critical Design Review presentation

Outgoing: Critical Design Package **Project maturity:** 20–35% **Engineering maturity:** 60%